EEEN20252 Lab Report

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Aero 2 ID	N/A		
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LAB 1

Task 1b: Record the rotor parameters estimated from the step response in Table 1.

Table 1. Rotor parameters

Parameter	Value
K	18.74
τ	0.174

Task 2b: Plot of comparison between estimated rotor model and physical Aero 2 model.

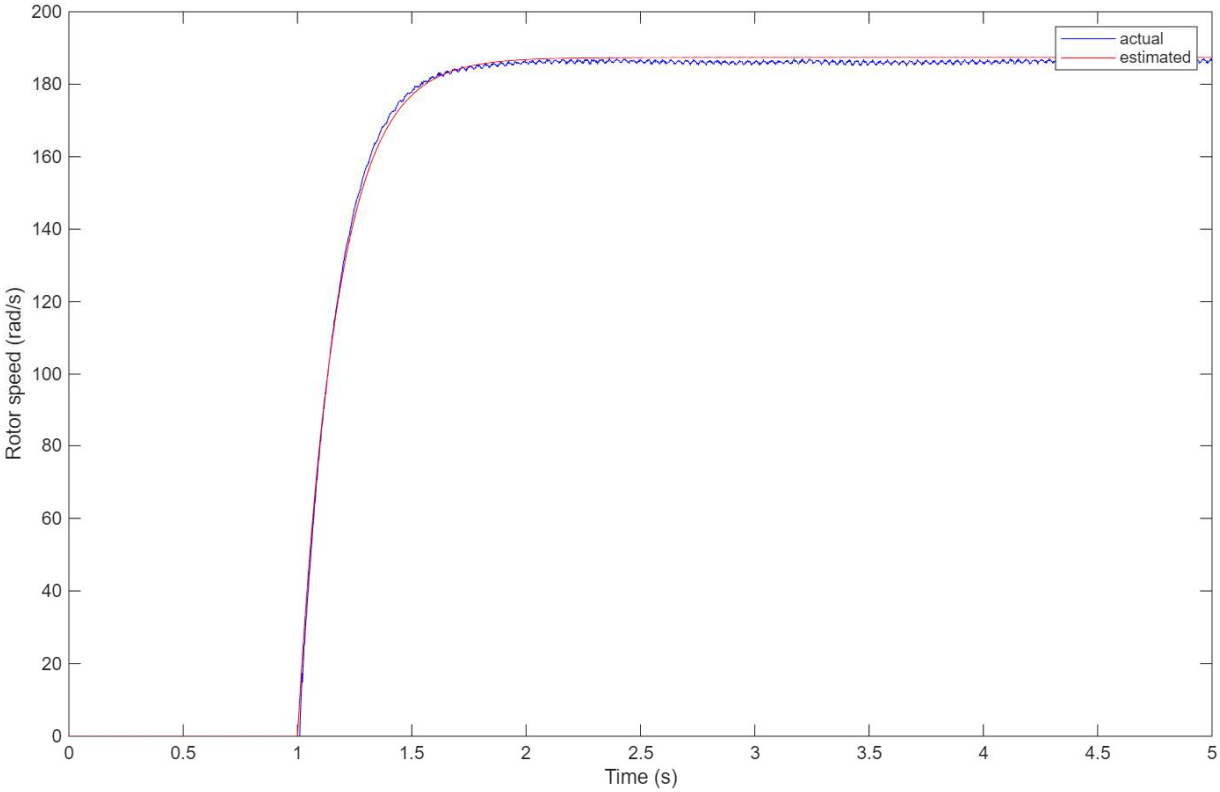


Figure 1: Step response of estimated rotor model versus actual Aero 2 model

Task 2c: Comments on the differences between the step response of the estimated model and the physical Aero 2 in terms of time constant and steady state value and state potential reasons for the differences.

Similarities:
The steady state value, response speed and shape of the curve is almost identical. This means our simulated model and the physical machine are running the exact same test with the exact same output.

Differences:
The actual curve has a lower steady state value compared to the estimated value. This might have happened because we used our own steady state gain and time constant that caused the actual experiment to be an almost identical copy of the simulated experiment but not a 1 to 1 copy.

The physical machine has noise compared to the estimated model. The estimated model can only run a simulation, and it doesn't simulate the fact that noise can cause our output to be different to the output of the physical machine test.

Task 3c: Record the pitch model parameters estimated from Task 3 in Table 2.

Table 2. Pitch model parameters

Parameter	Value
Damping ratio ζ	0.0259
Natural frequency ω_n (rad/s)	0.9935
Viscosity constant D_{vp} (N.m/(rad/s))	0.0012
Stiffness constant K_{sp} (N.m/rad)	0.0228
Thrust force gain K_{pp} (N/(rad/s))	0.0003447

Task 4b: Plot showing comparison between estimated pitch model and physical Aero 2 pitch response.

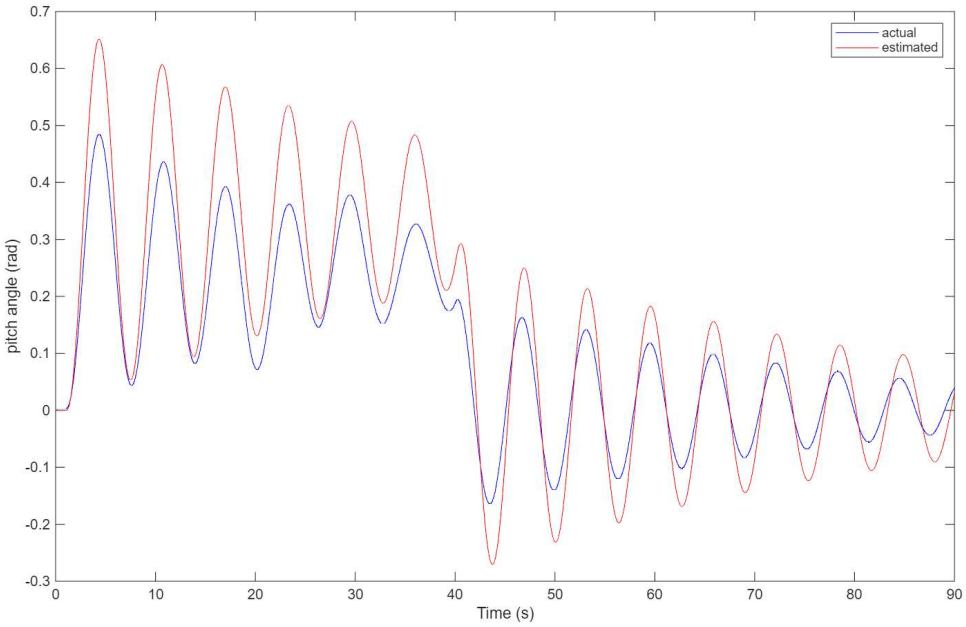


Figure 2: Pitch response of estimated pitch model versus actual Aero 2 model

Task 4c: Comments on whether estimated pitch model matches physical Aero 2 pitch response and why.

The estimated pitch model generally matches the physical Aero 2 pitch response because both responses show similar oscillation behavior, damping trend and settling characteristics. However, there are still small differences between the simulated and physical responses. The physical Aero 2 response contains more noise and slight variations in oscillation amplitude due to unmodelled nonlinear effects, friction, aerodynamic disturbances, sensor noise and parameter estimation errors. Overall, the estimated model is sufficiently accurate for controller design purposes.

LAB 2

Task 5a: Comments on how increasing/decreasing proportional gain affects the response in terms of steady state error and rise time.

The steady state error is the difference between the estimated and actual value. As we increase k_p further, the estimated steady state value increases. So, the settling point of the steady state value reaches closer to the reference point as we increase k_p. This shows the steady state value error decreases as we increase k_p. The rise time also decreases by microsecond values with each k_p increment, which allows the steady state value to be reached quicker. So, as we increase k_p, steady state error and rise time decreases.

Task 5b: Speed response plot with Proportional controller and estimation of % steady-state error.

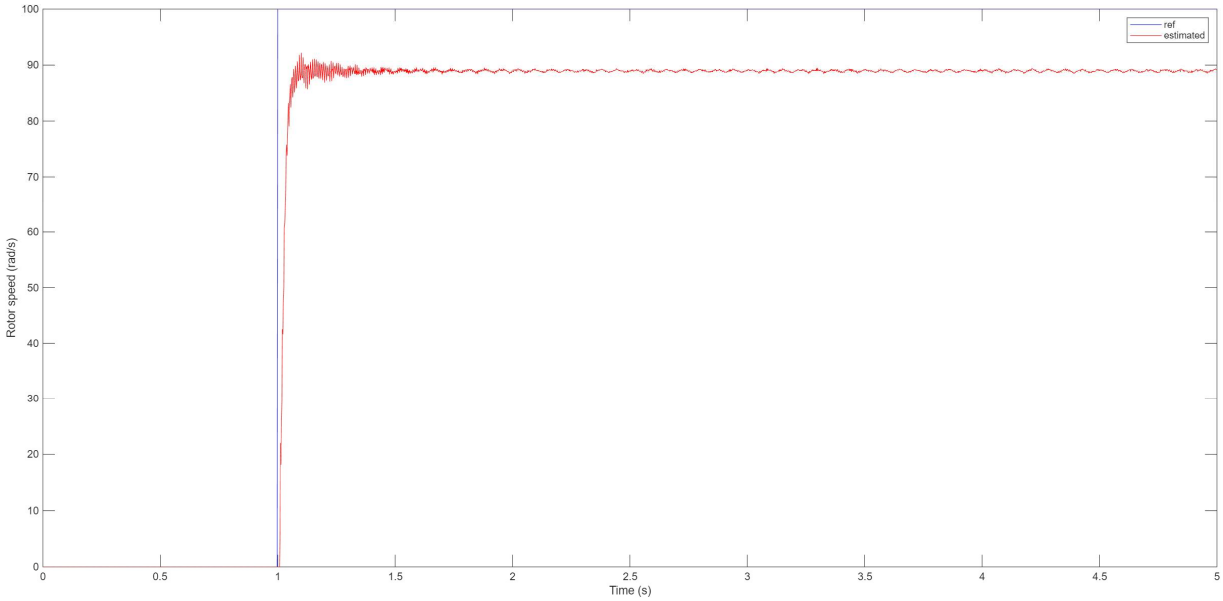


Figure 3: Rotor speed response with Proportional control only

Percentage Steady-state error (%) = $\frac{reference\ value\ -\ \ steady{-}state\ vakye}{reference\ value} * 100$ (1)

Using (1) with the measured steady-state speed at 89 rad/s and reference value at 100 rad/s the Percentage Steady-state error: $\frac{100-89}{100} * 100 = 11\ \%$

Task 6a: Computed desired damping ration and natural frequency based on performance requirements.

Table 3: Desired damping ratio and natural frequency

Parameters	Value
Damping ratio ζ	0.5169
Natural frequency ω_n (rad/s)	8.4367

Task 6b: Computation of proportional and integral gains for rotor control.

Proportional gain:

$$k_{pr} = \frac{2*\zeta*w_n - a}{b} = \frac{2*0.5169*8.4367 \quad .7471}{107.7} = 0.02762 \qquad (2)$$

Integral gain:

$$T_{ir} = \frac{b*k_{pr}}{w^2{}_n} = \frac{107.7*0.02762}{(8.4367)^2} = 0.04179 \qquad (3)$$

We fine-tuned our values to be closer to the theoretical peak overshoot and peak time. We kept the time integral constant but varied k_p until the value matched the theoretical values. Initially, the peak overshoot was 16.2% and the peak time was 0.39 seconds. After fine-tuning, the proportional gain k_{pr} is reduced to 0.025 and integral gainT_{ir} is kept the same. The peak overshoot and peak time are:

$$P_o = 15.37\%$$

Peak time = 0.41 s

Task 7a: Plot showing desired and measured rotor speed using Proportional-Integral control.

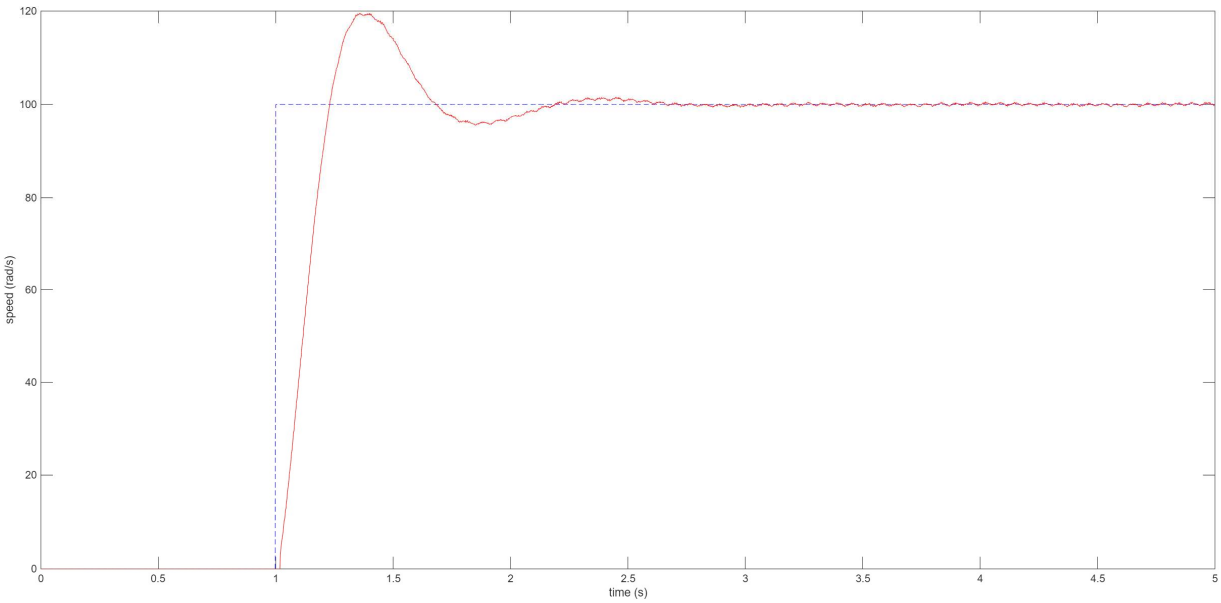


Figure 4: Rotor speed response with Proportional-Integral control

Peak overshoot: 19.22%
Peak time: 0.386 s
Task 7b: Comments on differences between the Matlab simulated responses and the responses from the Aero 2 hardware in terms of overshoot and rise time and state potential reasons for the differences.

The Aero 2 response has a larger overshoot and had a shorter response time compared to the MATLAB simulation. This could be due to MATLAB simulating a first order rotor model whereas the real hardware Aero 2 included real effects that are not modelled in MATLAB such as sensor noise, friction, drag, and delay.

Task 7c: Comments on effects of integral action after comparing P-only (Task 6) versus PI control (Task 7).

Compared with the P-only controller, the PI controller significantly reduced the steady-state error. The integral action accumulates the error over time and continuously adjusts the control input until the measured rotor speed matches the reference speed. However, adding integral action also increased the overshoot slightly and made the response more oscillatory.

Task 8b: Trial and error tuning of PIDγ controller for pitch angle.
Final PID gains:
Proportional gain: 1000
Integral gain: 1
Derivative gain: 1.05
γ value: 0.1

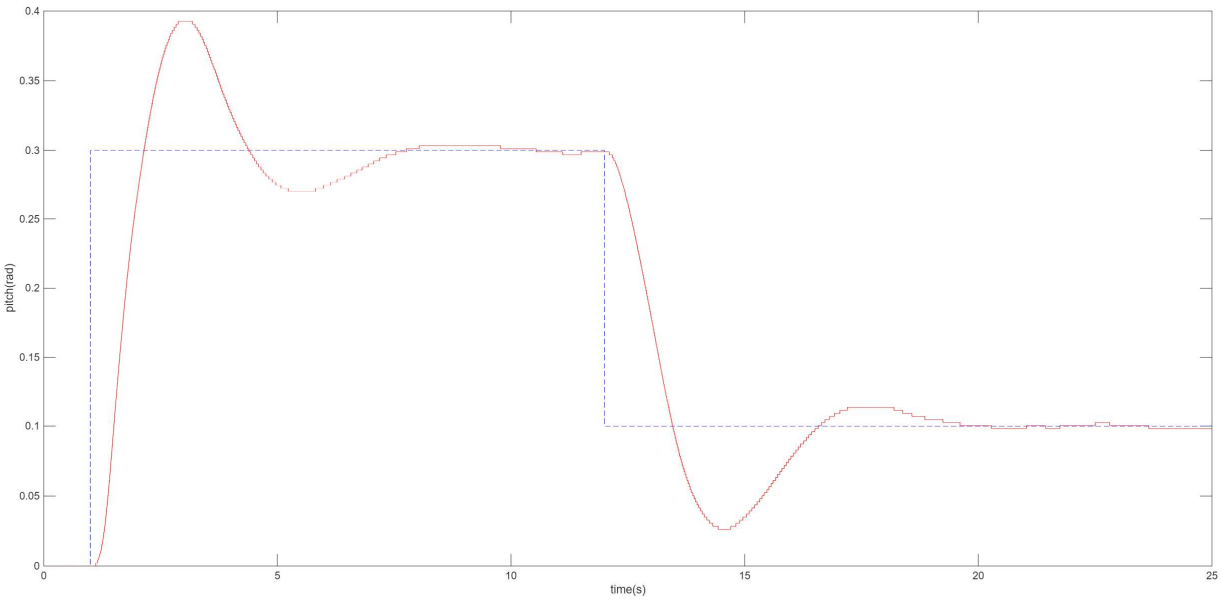


Figure 5: Pitch angle response with Proportional-Integral-Derivative-Gamma control

Task 9a: Transfer function from input voltage to rotor speed.

Starting from the motor voltage and rotor shaft equations:

$$v_m(t) = R_m i_m(t) + L_m \frac{di_m(t)}{dt} + k_m \omega_m(t) \quad (4)$$

$$J_{eq} \frac{d\omega_m(t)}{dt} = k_t i_m(t) - k_{dd} \omega_m(t) \quad (5)$$

Performing Laplace transform on (4) and (5):

$$V_m(s) = (R_m + L_m s) I_m(s) + k_m \Omega_m(s) \quad (6)$$

$$J_{eq} s \Omega_m(s) = k_t I_m(s) - k_{dd} \Omega_m(s) \quad (7)$$

Rearrange (7) to get $I_m(s)$ alone:

$$I_m(s) = \frac{J_{eq} s + k_{dd}}{k_t} \Omega_m(s) \quad (8)$$

Substitute (8) into (6):

$$V_m(s) = (R_m + L_m s) \left(\frac{J_{eq} s + k_{dd}}{k_t} \Omega_m(s) \right) + k_m \Omega_m(s) \quad (9)$$

Factor out $\Omega_m(s)$ and rearrange the formula to get the transfer function:

$$G(s) = \frac{\Omega_m(s)}{V_m(s)} = \frac{1}{\frac{(R_m + L_m s)(J_{eq} s + k_{dd})}{k_t} + k_m} \quad (10)$$

Multiplying the top and bottom of (10) by k_t and expand the denominator:

$$G(s) = \frac{\Omega_m(s)}{V_m(s)} = \frac{k_t}{L_m J_{eq} s^2 + (L_m k_{dd} + R_m J_{eq}) s + (R_m k_{dd} + k_t k_m)} \quad (11)$$

Task 9b: Transfer function from rotor speed to pitch angle.

Starting from the pitch equation:

$$J_{pp} \ddot{\theta}(t) + D_{vp} \dot{\theta}(t) + K_{sp} \theta(t) = K_{pp} D_t \omega_m(t) \quad (12)$$

Performing Laplace transform on (12):

$$J_{pp} s^2 \Theta(s) + D_{vp} s \Theta(s) + K_{sp} \Theta(s) = K_{pp} D_t \Omega_m(s) \quad (13)$$

Factor out $\Theta(s)$ and rearrange the formula to get the pitch transfer function:

$$\frac{\Theta(s)}{\Omega_m(s)} = \frac{K_{pp} D_t}{J_{pp} s^2 + D_{vp} s + K_{sp}} \quad (14)$$

To compare this with the standard second-order form, divide the numerator and denominator by J_{pp}:

$$\frac{\Theta(s)}{\Omega_m(s)} = \frac{\frac{K_{pp} D_t}{J_{pp}}}{s^2 + \frac{D_{vp}}{J_{pp}} s + \frac{K_{sp}}{J_{pp}}} \quad (15)$$

The standard second-order denominator is:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 \quad (16)$$

Comparing the coefficients of the denominator of (15) with (16) gives us our final equations:

$$\frac{D_{vp}}{J_{pp}} = 2\zeta\omega_n \implies D_{vp} = 2J_{pp}\zeta\omega_n \quad (17)$$

$$\frac{K_{sp}}{J_{pp}} = \omega_n^2 \implies K_{sp} = J_{pp}\omega_n^2 \quad (18)$$

Task 9c: Closed loop transfer function from reference speed to error when using Proportional controller.

For the P-only controller:

$$V(s) = k_p E(s) \tag{19}$$

Transfer function for the rotor model used is:

$$G(s) = \frac{\Omega_m(s)}{V(s)} = \frac{b}{s+a} \tag{20}$$

Rearrange (20) and substitute (19) into it:

$$\Omega_m(s) = G(s)k_p E(s) \tag{21}$$

The error equation is:

$$E(s) = R(s) - \Omega_m(s) \tag{22}$$

Substitute (21) into (22):

$$E(s) = R(s) - G(s)k_p E(s) \tag{23}$$

Move all $E(s)$ terms to one side and factor it out:

$$E(s)(1 + k_p G(s)) = R(s) \text{ ==> } \frac{E(s)}{R(s)} = \frac{1}{1+k_p G(s)} \tag{24}$$

Substitute right hand side of (20) into (24):

$$\frac{E(s)}{R(s)} = \frac{1}{1+k_p \frac{b}{s+a}} \tag{25}$$

Multiplying numerator and denominator by $(s + a)$ gives the closed-loop transfer function from reference speed to error:

$$\frac{E(s)}{R(s)} = \frac{s+a}{s+a+k_p b} \tag{26}$$

This shows that with P-only control, the error is reduced when $k_p$is increased, but it is not completely removed. This matches what we saw in the P-only experiment, where there was still steady-state error.

Task 10: Summary of teamwork and reflection on team performance.

Overall, our teamwork went smoothly. We communicated continuously throughout both lab sessions and ensuring all members understood each task before moving on. We successfully completed all bath labs within the allocated time, often with time to spare. One strength of our teamwork was our approach to uncertainty. When a member did not fully understand a task, we paused, discussed the issue together, and then confirmed our understanding with GTAs, lecturers or other teams. This prevented mistakes early on and improved the quality of our responses. Additionally, when one teammate proposed a solution, they actively asked for others' opinions before proceeding. This collaborative decision-making ensured that our answers to questions and queries were well-reasoned. We deliberately chose not to assign fixed roles, such as one person operating the simulation and another editing the report. Instead, we took turns running the hardware, logging data, and writing up observations. This allowed each of us to gain a complete understanding of the entire lab. However, there are areas for improvement. Our communication within the group was sometimes overshadowed by communication with other groups. We occasionally spent too long asking neighbouring teams how they approached a problem, rather than trusting our initial approach, causing further distractions and delaying progress. In future labs, we will agree to keep the distractions to a minimum and focus on the task at hand when consulting others. Overall, our teamwork was effective, productive, and contributed positively to the success of the lab.